



Customer: EDGEN MURRAY CORP

P.O. No. STL328793 / 4501048440

S.O. No. 523777

[illegible]



An AFG Holdings Company

Material Test Report

Heat Code: NU7

Ameriforge Ltd

13770 Industrial Rd. Houston, TX 77015

Sales: (713) 393-4200

ISO 9001:2015 Certified

NOV-R&M ENERGY SYS	15278	PO: 4057939	Sales Order: 202546	Line: 1
500 CONROE PARK WEST DR		Item Code: 9999H60039-0031R	Qty Shipped: 265	
CONROE, TX 77303		Item Desc: 4.00 150-900 CAP A350LF2 - 17518063-001 NRM CDFRG		
		Mill Heat: A256488		
Spec: ASME SA350/SA350M(18) LF2 CL1 & NOV 61-0072 Rev 01 (14) (Caps)				

Element	(%wt)	Ladle	Product	EPCRA	CAS#	Element	(%wt)	Ladle	Product	EPCRA	CAS#
C Carbon	0.17	/				Mo Molybdenum	0.03	/			
Mn Manganese	1.12	/		Y	7439-96-5	V Vanadium	0.03	/			
P Phosphorous	0.008	/				Cb Columbium	0.00	/			
S Sulphur	0.01	/				Al Aluminum	0.02	/			
Si Silicon	0.24	/				C.E.	0.40	/			
Cu Copper	0.15	/		Y	7440-50-8	CuNiCrMoV	0.35	/			
Ni Nickel	0.06	/		Y	7440-02-0	CrMo	0.11	/			
Cr Chromium	0.08	/		Y	7440-47-3						

Mechanical Testing		Heat Treat	
Test Lab	AFG	Norm. Temp. (F)	1675
Test Bar Size	4" 150-900 CAP. TP AT 1/2T	Norm. (Hrs)	2.5
HBW	137 - 137	Norm. Cooling Type	AIR
Elg (%)	31		
RA (%)	59	Other	
Tensile Specimen Size (in)	.245	Supplier	JADE STERLING
Tensile (ksi)	77	Country of Melt	USA
Yield (ksi)	50	Country of Forging	USA
Gauge Length (in)	1	Vacuum Degassed	Y
Charpy Temp (F)	-50		
Charpy (ft lbs)	44 - 61 - 72		
Avg Energy 3 Specimens (ft)	59		
Min Energy 1 Specimens (ft)	44		
% Shear	30 - 40 - 50		
Lat. Exp. (in)	.039 - .058 - .063		

PO: 4057939
PN: 17518063-001
HT: NU7

4-CAP. 265
51063708

The recording of false, fictitious, or fraudulent statement or entries on this document may be punishable as a felony under federal statute.

EPCRA Supplier Notification: This product may contain one or more toxic chemicals subject to the reporting requirements of Section 313 of the Emergency Planning and Community Right-to-Know Act (Title III of the Superfund Amendments and Reauthorization Act of 1986) and 40 C.F.R. Part 372. Potentially reportable chemicals are indicated with a checkmark in the "EPCRA" column and a Chemical Abstract Services (CAS) registry number is provided for each such chemical in addition to the percent by weight of the chemical present in this product. It is your responsibility alone to determine whether your facility is required to submit a Toxic Release Inventory Report under EPCRA Section 313.

Certification No.: 1693603
Certification Date: 3/30/2026
Issued By: Michelle Simien

Approved By:

This report is issued in compliance with the requirements of EN10204 3.1 / ISO 10474 3.1.b

Material Test Report

Heat Code: NU7

Ameriforge Ltd

13770 Industrial Rd. Houston, TX 77015

ISO 9001:2015 Certified

Sales: (713) 393-4200

NOV-R&M ENERGY SYS	15278	PO: 4057939	Sales Order: 202546	Line: 1
500 CONROE PARK WEST DR		Item Code: 9999H60039-0031R	Qty Shipped: 265	
CONROE, TX 77303		Item Desc: 4.00 150-900 CAP A350LF2 - 17518063-001 NRM CDFRG		
		Mill Heat: A256488		
Spec: ASME SA350/SA350M(18) LF2 CL1 & NOV 61-0072 Rev 01 (14) (Caps)				

Material is Fully Killed; Fine Grain Practice in accordance with A-29/SA-20 Section 8.3.

TAG H60039

Product compliant with NACE MR0175/ISO 15156-2:2021, Annex A (A.2.1.2, A2.1.3).

Product compliant with NACE MR0103/ISO 17945:2015, Section 13.1.1 and 13.1.3.

- * No Weld repair performed.
- * Ladle Chemical Analysis results are reported from the raw material suppliers MTR.
- * Tensile taken from a round specimen. Measurement of elongation after fracture.
- * Yield strength was determined using the .2% offset method, unless specified.
- * All material supplied under this order is certified to be free of mercury contamination and no mercury bearing equipment was used in manufacturing, fabrication or testing.
- * When reported on MTR: Tensile Testing Per ASTM E8; Brinell Hardness Per ASTM E10; Charpy Testing Per ASTM E23
- * AF Global manufactures materials according to a certified Quality Management System conforming to ISO 9001:2015 and PED certified to 2014/68/EU, Annex 1, Section 4.3.
- Unless otherwise noted on PO, Dimensions of all standard pipe flanges and flange fittings (NPS 1/2 to NPS 24) per ASME B16.5-2020, large diameter steel flanges (NPS 26 to 60) per ASME B16.47-2020, Orifice flanges per ASME B16.36-2020.
- * AFGlobal uses the latest edition of ASTM E29 (Standard practice for using significant digits in test data to determine conformance with specifications), to claim conformance or nonconformance to a specification. When a accept/reject determination is required AFGlobal uses a test uncertainty ratio (TUR) of 1:1 or higher as acceptable.
- * DNV PED Certificate No 10000425664-PA-DNV-USA Valid thru December 03, 2026.

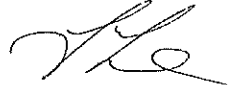
The recording of false, fictitious, or fraudulent statement or entries on this document may be punishable as a felony under federal statute.

EPCRA Supplier Notification: This product may contain one or more toxic chemicals subject to the reporting requirements of Section 313 of the Emergency Planning and Community Right-to-Know Act (Title III of the Superfund Amendments and Reauthorization Act of 1986) and 40 C.F.R. Part 372. Potentially reportable chemicals are indicated with a checkmark in the "EPCRA" column and a Chemical Abstract Services (CAS) registry number is provided for each such chemical in addition to the percent by weight of the chemical present in this product. It is your responsibility alone to determine whether your facility is required to submit a Toxic Release Inventory Report under EPCRA Section 313.

Certification No.: 1693603

Certification Date: 3/30/2026

Issued By: Michelle Simien


Dusty Bannon - Quality Analyst

This report is issued in compliance with the requirements of EN10204 3.1 / ISO 10474 3.1.b

Page 2 of 2

NU7

NOV-R&M ENERGY SYS	15278	PO: 4059803	Sales Order: 202936	Line: 1
500 CONROE PARK WEST DR		Item Code: 9999H61985-0031R	Qty Shipped: 333	
CONROE, TX 77303		Item Desc: 4.00 150-900 HUB A350LF2 - 17518507-001 NRM CDFRG		
		Mill Heat: A256488		
Spec: ASME SA350/SA350M(24) LF2 CL1 & NOV 61-0072 (14) Rev 01 (Hubs)				

Element	(%wt)	Ladle	Product	EPCRA	CAS#	Element	(%wt)	Ladle	Product	EPCRA	CAS#
C	Carbon	0.17	0.16			Mo	Molybdenum	0.03	0.02		
Mn	Manganese	1.12	1.12	Y	7439-96-5	V	Vanadium	0.03	0.02		
P	Phosphorous	0.008	0.006			Cb	Columbium	0.00	0.00		
S	Sulphur	0.01	0.01			Al	Aluminum	0.02	0.02		
Si	Silicon	0.24	0.26			C.E.		0.40			
Cu	Copper	0.15	0.14	Y	7440-50-8	CuNiCrMoV		0.35			
Ni	Nickel	0.06	0.05	Y	7440-02-0	CrMo		0.11			
Cr	Chromium	0.08	0.08	Y	7440-47-3						

4" Hubs

Mechanical Testing		Heat Treat	
Test Lab	AFG	Norm. Temp. (F)	1675
Test Bar Size	TP AT 1/2T LONG.	Norm. (Hrs)	2.5
HBW	137 - 137	Norm. Cooling Type	AIR
Elg (%)	33		
RA (%)	62		
Tensile Specimen Size (in)	.245	Other	
Tensile (ksi)	77	Supplier	JADE STERLING
Yield (ksi)	49	Country of Melt	USA
Gauge Length (in)	1	Country of Forging	USA
Specimen Orientation	LONGITUDINAL	Vacuum Degassed	Y
LCVN Charpy Temp. (F)	-50	Made to Fine Grain Practice	YES
LCVN (ft lbs)	60 - 80 - 71		
LAvg 3(ft-lbs)	70		
L Min (ft-lbs)	60		
% Shear	40 - 50 - 50		
Lat. Exp.	.057 - .072 - .068		

3/17/26

The recording of false, fictitious, or fraudulent statement or entries on this document may be punishable as a felony under federal statute.

EPCRA Supplier Notification: This product may contain one or more toxic chemicals subject to the reporting requirements of Section 313 of the Emergency Planning and Community Right-to-Know Act (Title III of the Superfund Amendments and Reauthorization Act of 1986) and 40 C.F.R. Part 372. Potentially reportable chemicals are indicated with a checkmark in the "EPCRA" column and a Chemical Abstract Services (CAS) registry number is provided for each such chemical in addition to the percent by weight of the chemical present in this product. It is your responsibility alone to determine whether your facility is required to submit a Toxic Release Inventory Report under EPCRA Section 313.

Certification No.: 1693151

Certification Date: 3/13/2026

Issued By: Terri Gibson

Page 1 of 2

This report is issued in compliance with the requirements of EN10204 3.1 / ISO 10474 3.1.b

Material Test Report

Heat Code: NU7

Ameriforge Ltd

13770 Industrial Rd. Houston, TX 77015

ISO 9001:2015 Certified

Sales: (713) 393-4200

NOV-R&M ENERGY SYS	15278	PO: 4059803	Sales Order: 202936	Line: 1
500 CONROE PARK WEST DR		Item Code: 9999H61985-0031R	Qty Shipped: 333	
CONROE, TX 77303		Item Desc: 4.00 150-900 HUB A350LF2 - 17518507-001 NRM CDFRG		
		Mill Heat: A256488		
Spec: ASME SA350/SA350M(24) LF2 CL1 & NOV 61-0072 (14) Rev 01 (Hubs)				

TAG H61985

Product compliant with NACE MR0175/ISO 15156-2:2021, Annex A (A.2.1.2, A2.1.3).
Product compliant with NACE MR0103/ISO 17945:2015, Section 13.1.1 and 13.1.3.

- * No Weld repair performed.
- * Ladle Chemical Analysis results are reported from the raw material suppliers MTR.
- * Tensile taken from a round specimen. Measurement of elongation after fracture.
- * Yield strength was determined using the .2% offset method, unless specified.
- * All material supplied under this order is certified to be free of mercury contamination and no mercury bearing equipment was used in manufacturing, fabrication or testing.
- * When reported on MTR: Tensile Testing Per ASTM E8; Brinell Hardness Per ASTM E10; Charpy Testing Per ASTM E23
- * AF Global manufactures materials according to a certified Quality Management System conforming to ISO 9001:2015 and PED certified to 2014/68/EU, Annex 1, Section 4.3.
- Unless otherwise noted on PO, Dimensions of all standard pipe flanges and flange fittings (NPS 1/2 to NPS 24) per ASME B16.5-2020, large diameter steel flanges (NPS 26 to 60) per ASME B16.47-2020, Orifice flanges per ASME B16.36-2020.
- * AFGlobal uses the latest edition of ASTM E29 (Standard practice for using significant digits in test data to determine conformance with specifications), to claim conformance or nonconformance to a specification. When a accept/reject determination is required AFGlobal uses a test uncertainty ratio (TUR) of 1:1 or higher as acceptable.
- * DNV PED Certificate No 10000425664-PA-DNV-USA Valid thru December 03, 2026.

The recording of false, fictitious, or fraudulent statement or entries on this document may be punishable as a felony under federal statute.

EPCRA Supplier Notification: This product may contain one or more toxic chemicals subject to the reporting requirements of Section 313 of the Emergency Planning and Community Right-to-Know Act (Title III of the Superfund Amendments and Reauthorization Act of 1986) and 40 C.F.R. Part 372. Potentially reportable chemicals are indicated with a checkmark in the "EPCRA" column and a Chemical Abstract Services (CAS) registry number is provided for each such chemical in addition to the percent by weight of the chemical present in this product. It is your responsibility alone to determine whether your facility is required to submit a Toxic Release Inventory Report under EPCRA Section 313.

Certification No.: 1693151
Certification Date: 3/13/2026
Issued By: Terri Gibson


Dusty Bannon - Quality Analyst

This report is issued in compliance with the requirements of EN10204 3.1 / ISO 10474 3.1.b